

Module 1

- 1. What is Database? Explaining the three schema architecture with a neat diagram.**
- 2. What are the advantages of using the DBMS approach? Explain**
- 3. Explain a) Data Dictionary b) Entity types c) Strong Entity d)Generalization..**
- 4. Explain the categories of Data Models.**
- 5. With a neat diagram, explain the components modules of DBMS.**
- 6. What are the responsibilities of DBA & Database Designer?**
- 7. Explain the types of end users with examples.**
- 8. Explain the characteristics of Database.**
- 9. Explain the different types of attributes in DBMS.**
- 10.Explain in detail about structural constraints.**
- 11.Demonstrate the concepts of a) Generalization. b) Specialization with examples.**
- 12. What is cardinality Ratio? What are the different types of cardinality ratio in a binary relationship?**

Module 2

- 1. Explain the need of Primary and Foreign Keys With suitable examples.**
- 2. Explain with sketch the different phases of Database design process.**
- 3. What is cardinality Ratio? What are the different types of cardinality ratio in a binary relationship?**
- 4. List out the notations for ER diagram and their meaning.**
- 5. What is a weak entity? Explain the role of partial key in design of weak entity type.**
- 6. Define the following terms with an example for each**
 - A . Candidate key**
 - B . Primary Key**
 - C. Super key**
 - D. Recursive relationship**

- E . Weak entity type
- F. Atomic attributes
- G. Participation role
- H. Entity
- I .Mapping Constraints
- J . Participation Constraints
- K .Strong Entity
- L . Recursive Relationships
- M. Equijoin and Natural join
- N. Rename

7. Consider the two tables T1 & T2

Table T1			Table T2		
P	Q	R	A	B	C
10	a	5	10	b	6
15	b	8	25	c	3
25	a	6	10	b	5

Show that result of the following operations

- i. $T1 \bowtie_{T1.P=T2.A} T2$
- ii. $T1 \bowtie_{T1.Q=T2.B} T2$
- iii. $T1 \bowtie_{T1.Q=T2.A} T2$
- iv. $T1 \bowtie_{T1.Q=T2.B} T2$
- v. $T1 \cup T2$

8. List the Characteristics of relation. Discuss each one.

9.Explain the Division Operator with an example. How can a division operator be implemented using other relational algebraic operators?

10. Consider the following three tables –SAILORS, RESERVES & BOATS having the following attributes.

SAILORS (Sid, Sname, Rating, Age)

RESERVES(Sid, Bid, Day) BOATS(Bid, Bname, Color)

Use the above schema & solve the queries in relational algebra

- a. Find names of sailors who reserved green boat**
- b. Find the colors of boats reserved by “Rajesh”**
- c. Find names of sailors who have reserved a red or a green boat.**
- d. Find the Sids of sailors with age over 20 who have not registered a red boat.**
- e. Find the name of the sailors who have reserved boats 103**
- f. Find the name of the sailors, who have reserved at least two boats**
- g. Retrieve the sailors' names who have reserved red green boats.**
- h. Retrieve the sailors names with age over 20 years and reserved black boat.**
- i. Retrieve the number of boats which are not reserved.**
- j. Retrieve the sailors names who have reserved green boats on Monday.**
- k. Retrieve the sailors names who is the oldest sailor with rating 10.**

11. Define different set operations in relational algebra. Give one example for each.

12. What do you mean by Integrity constraint w.r.t database? Explain entity integrity & Referential Integrity constraint.

13. List and explain Aggregate functions commonly used in relation algebra.

14. Explain violation during Update operations.

15. What is join? Explain the different types of joins.

16. Explain Cross product with example.

Module 3

- 1. What is the need for normalization? Discuss the informal guidelines for relational schema design.**
- 2. Define 1NF,2NF,3NF,BCNF,4NF,5NF with examples.**
- 3. Discuss insertion,deletion and modification anomalies. Why are they considered bad? Illustrate with examples.**
- 4. Illustrate the following with examples.**
 - a. Datatypes in SQL**
 - b. Substring pattern matching in SQL**
- 5. Explain multivalued dependency and 4NF with example.**
- 6. Explain Triggers and stored procedures with examples.**
- 7. Outlines the constraints in SQL.**
- 8. Explain the following with examples.**
 - a. Correlated and Non Correlated Nested Queries.**
 - b. EXISTS and NOT EXIST.**
 - c. UNIQUE & DISTINCT.**
- 9. Define Database stored procedure. Explain creating and calling of stored procedures with examples.**
- 10.What are views ? Explain the specification and implementation of views in SQL**
- 11. How are triggers and assertions specified in SQL. Explain with examples.**
- 12.EMPLOYEE(EmpID, EmpName, DeptID, DeptName)**

Assume the following functional dependencies:

EmpID → EmpName,

DeptID DeptID → DeptName

Task: Apply normalization and convert the relation into 1NF, 2NF, and 3NF.

Show the final set of relations.

Answer:

First Normal Form (1NF)

The relation contains only atomic values, so it is already in **1NF**.

EmpID	EmpName	DeptID	DeptName
E01	Alice	D01	HR
E02	Bob	D02	Finance
E03	Charlie	D01	HR

Second Normal Form (2NF)

Since the primary key is a **single attribute (EmpID)**, there are **no partial dependencies**.

Therefore, the relation is already in **2NF**.

2NF Relation:

EMPLOYEE(EmpID, EmpName, DeptID, DeptName)

Third Normal Form (3NF)

There is a **transitive dependency**:

- EmpID $\rightarrow$ DeptID
- DeptID $\rightarrow$ DeptName

Therefore:

EmpID $\rightarrow$ DeptName (Transitive Dependency)

To remove it, decompose the relation as follows:

EmpID	EmpName	DeptID
E01	Alice	D01
E02	Bob	D02
E03	Charlie	D01

DEPARTMENT

DeptID	DeptName
D01	HR
D02	Finance

Final 3NF Relations

1. **EMPLOYEE(EmpID, EmpName, DeptID)**
 - Primary Key: EmpID
2. **DEPARTMENT(DeptID, DeptName)**
 - Primary Key: DeptID

Conclusion

- **1NF:** Original relation
- **2NF:** No change (single-attribute primary key)
- **3NF:** Decomposed into EMPLOYEE and DEPARTMENT to remove transitive dependency.

13) Illustrate the normalisation process for the following table till 3NF.

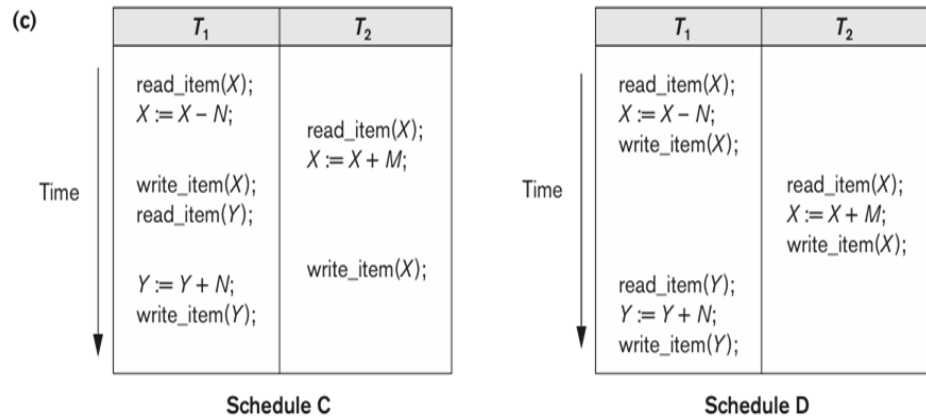
Student_ID	Student_Name	Course_Code	Course_Name	Faculty_ID	Faculty_Name
S001	Ravi	C01	DBMS	F01	Kumar
S001	Ravi	C02	Java	F02	Anita
S002	Priya	C01	DBMS	F01	Kumar
S002	Priya	C03	Python	F03	Raj
S003	Kiran	C02	Java	F02	Anita

Functional Dependencies

1. **Student_ID → Student_Name**
2. **Course_Code → Course_Name, Faculty_ID**
3. **Faculty_ID → Faculty_Name**
4. **(Student_ID, Course_Code)** is the Candidate Key

Module 4

1. Explain the need of Concurrency Control in transaction with example.
2. Discuss the ACID properties of transactional database.
3. Discuss the properties of a transaction with state transition diagram.
4. Explain different types of Failures.
5. What is Schedule? Explain conflict serializable schedule with example?
6. What are the anomalies due to interleaved execution of transactions? Explain with examples.
7. With a neat diagram, explain the typical states that a transaction goes through during the execution.
8. Describe Triggers and Assertions in SQL with syntax and example.
9. Projects (*pid, pname, budget*)
Employees (*eid, ename, salary*)
Works_On (*eid, pid*)
Departments (*did, dname*)
 - a) For each employee working on more than three projects, find the employee ID and the maximum budget among the projects assigned to them.
 - b) Print the name and salary of every employee who is not assigned to any project.
 - c) Find the names of employees who work on at least one project with a budget greater than ₹10,00,000.
 - d) Find employees whose name starts with 'K'.
10. Illustrate with precedence graph, which of following schedules is conflict serializable.



Module 5

1. List and explain the four major categories of NOSQL system.
2. Discuss the following MongoDB CRUD operations with their syntax and suitable examples.
3. Briefly discuss about Neo4j data model.
4. Explain the characteristics of NoSQL related to distributed databases and systems.
5. Explain CAP Theorem.
6. Briefly discuss the two-phase locking techniques for concurrency control.